

Nicolae Ungur

(425) 240-1837 | nungur@cs.washington.edu | github.com/x3rdev | linkedin.com/in/nicolae-ungur

Education

University of Washington - Seattle

Expected Graduation: 2028

- B.S. Computer Science, B.A. Mathematics, GPA: 3.9
- Relevant coursework: Distributed Systems, Machine Learning, Data Structures & Parallelism, Advanced Linear Algebra.

Experience

Software Development Engineering Intern - Amazon

June 2026 - September 2026

- Member of AWS Lambda Generic Pollers team.
- Designed routing logic to decouple event source consumption from destination handling, enabling flexible endpoint configuration.
- Deployed a CI/CD pipeline on AWS using CDK, provisioning Lambda functions and MSK (Kafka) clusters for automated testing and release.

Freelance Software Developer

Oct 2024 - Present

- Designed and deployed a production e-commerce platform (React + Spring Boot) implementing authentication, payment processing and order workflows.
- Deployed and monitored backend services on AWS, improving site reliability and reducing manual downtime.
- Delivered multiple client projects concurrently, handling implementation and post-launch fixes without missed deadlines.

Undergraduate Teaching Assistant - University of Washington

Sep 2025 - Present

- Guided students in diagnosing runtime errors and logical bugs by reading stack traces and troubleshooting code.
- Achieved a 4.93/5 student evaluation rating, with feedback highlighting clear communication and effectiveness in debugging assistance.
- Collaborated with course staff to identify recurring failure patterns and refine instructional material.

Skills

Languages: Java, C++, C, Python, JavaScript/TypeScript

Backend & Systems: Spring Boot, Node.js, REST APIs, TCP/IP networking, SQL

Frameworks & Tools: Git, Linux/Unix Systems, JUnit, Gradle, AWS

Projects

Send File P2P

github.com/x3rdev/send_file_p2p

- Designed and implemented a high-performance peer-to-peer file transfer system in C++ using low-level socket programming with minimal server coordination.
- Implemented UDP hole punching to establish direct client-to-client connectivity behind NATs, avoiding centralized data relays.
- Developed multithreaded client logic to compress and transfer files efficiently under concurrent workloads.
- Improved transfer throughput and resilience through optimized error handling for unreliable network conditions.

OpenFrontMind

github.com/x3rdev/openfrontmind

- Trained a PPO agent in PyTorch for real time strategy game with 93% win rate against multiple opponents over 100+ hours of validation.
- Designed a CNN + attention-based entity encoder policy network that pools variable-sized inputs into fixed-size representation via attention pooling.
- Cut model size 13x (5.3M to 414K parameters) and reached 1,890 simulated game-steps per second across 32 parallel environments.
- Implemented self-imitation learning to reinforce winning behavior, breaking a stalled multi-opponent policy out of a 0.2% win-rate plateau to 87% within 672 training updates.

Minecraft Addon

github.com/x3rdev/MekanismTurrets

- Built and maintained a high-traffic open-source plugin with 1.5M+ downloads, ensuring compatibility across multiple Minecraft versions.
- Designed performance-critical game logic under tight memory and CPU constraints, avoiding frame drops during gameplay.
- Debugged post release issues by reviewing GitHub bug reports and coordinating with external contributors.